

CE 610: Analysis and Design for Earthquake Resistant Structures

Course Code	CE 610
Course Name	Analysis and Design for Earthquake Resistant Structures
L-T-P-C	3-0-0-3
Prerequisites	Structural Dynamics with Application to Earthquake Engineering (CE 511) or Equivalent
Intended for	M.Tech. (Structural Engineering); UG (4th year), PG (M.S., Ph.D.)
Distribution	Elective (UG/PG)/ Specialization Elective (M.Tech. - Structural Engineering)
Approval	16th Senate (Apr 2018), in the M.Tech Structural Engineering curriculum; 27th BoA (Nov 2018); 19th Senate (Feb 2019)

Course Contents

- **Basic Concepts:** Behavior of structures and structural components under earthquake loading; Introduction to terminology; Seismic inputs to structures; Examples of earthquake resistant structures and their behavior during actual earthquakes. [4 Lectures]
- **Seismic Design Philosophy:** Historical development of earthquake resistant design philosophy; Concept of strength overstrength and ductility; Concept of equal displacement and equal energy principles, capacity design. [4 Lectures]
- **Modeling and Analysis of Buildings:** Equivalent static analysis; Response spectrum analysis; Mode superposition method; Time history analysis; Non-linear analysis consideration; Pushover analysis; Modeling concept of reinforced concrete building; Consideration of irregularities in seismic design of buildings; Introduction to modeling and analysis of reinforced masonry buildings; Simulation of structures using software for seismic analysis. [12 Lectures]
- **Seismic Design of Building Components:** Earthquake resistant properties of reinforced concrete; Seismic behavior and design of linear reinforced concrete elements; Seismic behavior of planar reinforced concrete elements; Code provisions; Detailing for ductility; Design guidelines specific for reinforced masonry building; Earthquake resistant design considerations for steel structures. Consideration for computer aided seismic design using software simulation. [14 Lectures]
- **Advanced Topics in Earthquake Resistant Design:** Performance based earthquake engineering; Introduction to earthquake resistant design for special structures; Retrofitting and strengthening of structures; Concept of base isolation technique and other seismic vibration control. [8 Lectures]

Text Books

1. Pauley T. and Priestley M.J.N., **Seismic Design of Reinforced Concrete and Masonry Buildings**, John-Wiley & Sons, 1992.
2. Datta T.K., **Seismic Analysis of Structures**, John Wiley & Sons (Asia) Pte Ltd, 2010.

References

1. Chopra A.K., **Dynamics of Structures- Theory and Application to Earthquake engineering**, 5th Edition, Pearson Education India, 2017.
2. Shrikhande M., Agrawal P., **Earthquake Resistant Design of Structures**, Prentice Hall India Learning Private Limited, 2006.
3. Taly N., **Design of Reinforced Masonry Structures**, 2nd Edition, McGraw-Hill Education, 2010.
4. Rai D.C. (2005), **IITK-GSDMA Guidelines for Structural use of Reinforced Masonry - Provisions with Commentary and Explanatory Examples**, IIT Kanpur, Gujarat State Disaster Mitigation Authority.
5. ATC-40, **Seismic Evaluation and Retrofit of Concrete Building** - Vol. I & II, Applied Technology Council, 1996.
6. AISC 341-10, **Seismic Provisions for Structural Steel Buildings**, American Institute of Steel Construction, 2010.
7. Naeim F., Kelly J.M., **Design of Seismic Isolated Structures: From Theory to Practice**, John Wiley & Sons, 1999.
8. IS 189J (Part I), **Criteria for Earthquake Resistant Design of Structures: Part I General Provisions and Buildings**, Bureau of Indian Standard, 2016.
9. IS 13920, **Ductile Design and Detailing of Reinforced Concrete Structures Subjected to Seismic Forces-Code of Practice**, Bureau of Indian Standard, 2016.